JIANFENG CHEN

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WORK EXPERIENCE

Meta Platforms *(formerly Facebook)* | **Research Scientist** Jul 2019 - Present
Mobile Data Frameworks (GraphQL) *Programming Language, Compiler, Product Infra*

Language Compiler

- Major contributor to revamp OSS GraphQL compiler **using Rust**, tailored for client-side implementations across Meta's family of applications, including Facebook, Instagram, Messenger, and Meta AI;
- Spearheaded prototyping and optimization of the GraphQL compilation process, reducing build time for GraphQL modules within the Facebook app **by 99%**, from 38 minutes to 17 seconds;
- Worked and promoted the fragment variable as a new feature to the GraphQL OSS specifications.

Developer Tools, IDE and Infra

- Led the development of GraphQL **tool chains**, delivering real-time query and schema validation tools;
- Key contributor to GraphQL **IDE plugins** for VS Code and Android Studio, enhancing the developer experience;
- Orchestrated a **Cloud Based CI/CD** Platform for mobile developers.

Code Analysis and Automation

- Implemented **static code analysis** (code indexing) to identify and remove unreachable modules in Kotlin, Swift, and Objective-C, resulting in over **US$10 million** in server power savings;
- **Automated code authoring** for strong-typed code by static analysis and Large Language Models (LLMs).

Mentoring and Invited Talks

- Provided mentorship to three interns and facilitated the onboarding process for three mid/senior-level engineers;
- Invited talk (Droidcon 2021): *Integrating custom language with android studio: GraphQL case study*;
- Invited talk (cdCon 2022): *GraphQL Dev Supporting Tools in Meta*.

EDUCATION

PhD in Computer Science Aug 2014 - May 2019
North Carolina State University, United States *GPA: 4.0/4.0*
Advisor: Dr. Tim Menzies (ACM Fellow, IEEE Fellow, ASE Fellow)
Dissertation: On the Value of Sampling and Pruning for Search-Based Software Engineering

Master in Computer Science *(En route)* Aug 2014 - Dec 2018
North Carolina State University, United States *GPA: 4.0/4.0*
Coursework: Automated SE | DevOps | Advanced AI | Algorithm Analysis | Data Mining | Data-to-Knowledge

BS in Computer Science Sep 2010 - May 2014
Shandong University, China *GPA: 89.93/100*
Coursework: Data Structure | OS | Networking | Database System | Numerical Analysis | Compilers

SKILLS AND INTERESTS

Language: Rust, Python, C++, Kotlin, JavaScript and SQL;
Data analysis tools: Scikit-learn, SciPy, Pandas, jMetal, Gephi;
DevOps tools: Jenkins, Ansible, Travis-CI, AWS Elasticsearch, S3, Docker, Redis.

INTERN AND RESEARCH EXPERIENCE

Spike: Predicting Breakdowns in Cloud Services Sep 2018- Apr 2019
Cooperation project with LexisNexis Legal & Professional

- Fetched cloud services monitoring data in microservices systems;
- Cleaned up the data and built machine learning models to predict the breakdowns 30 minutes ahead;
- Got a alarm system for service spikes with recalls and precision of 75% or above;
- Published results on Symposium on the Foundations of Software Engineering 2019 (Industrial track).

Cross-Stan: Embedding Bayesian Modeling in Python and C++ Programs May 2018- Aug 2018
Internship at Facebook (Machine learning experience group)

- Created the tool *CxStan* to embed Bayesian Modeling engine, Stan, into Python and C++ programs;
- Built a hierarchical framework to accelerate the Monte Carlo Sampling for the Stan modeling;
- Integrated the tool into the buck build tool, making the Stan engine as a service;
- Applied the *CxStan* to some Facebook services' traffic policing.

Automated Configurations for Cloud-based Workflows May 2017 - Aug 2017
North Carolina State University

- Presented a novel stochastic method for rapidly configuration cloud-based workflows;
- Automatically deployed the workflow with more than 500 sub-tasks to AWS platform. Save up to 30% economy cost within specific deadline requirement, compared to default greedy deployment policy in AWS.

LACE Data Privatization Tools and its Application Aug 2016 - Nov 2016
NSA funded project in RAISE Lab

- Distributed a data anonymization package in Python , tested package via Travis-CI;
- Applied my package to education and medical data sets. Evaluate data set utility through supervised learning.

Fast Principal-component-analysis (F-PCA) Method for Flight Status Log May 2016 - Aug 2016
Google Summer of Code program 2016

- Accepted by Google `GSoC2016` program among 18,981 applicants **(accept rate: 6%)**;
- Hierarchical clustering a dataset(flight status log) with more than 20M entries top-down and bottom-up. Create a PCA-like dimension reduction algorithm and speed it up by spark. Compared my own algorithm with PCA.

Sampling vs. Searching in Search-based SE Dec 2014 - Aug 2017
North Carolina State University

- Created a sampling technique to solve the software product line problem. Deployed the algorithm into platform LSF by MPI; implemented a job schedule engine. Reduced the experiment time from 2 months CPU hrs into 11.5 hrs.
- Modeled the Linux Kernel modules in CNF sets. Created the decision tree surrogate model. By combining SAT solvers, found a way to configure large software systems 2000x faster. Published the results in TSE.

PUBLICATIONS

[1] Xueqi Yang, **Jianfeng Chen**, Rahul Yedida, Zhe Yu, and Tim Menzies. "Learning to recognize actionable static code warnings (is intrinsically easy)." Empirical Software Engineering (2021)

[2] Shu Rui, Tianpei Xia, **Jianfeng Chen**, Laurie Williams, and Tim Menzies. "How to better distinguish security bug reports (using dual hyperparameter optimization)." Empirical Software Engineering (2021)

[3] **Jianfeng Chen**, Joymallya Chakraborty, Philip Clark, Kevin Haverlock, Snehit Cherian and Tim Menzies. "Predicting Breakdowns in Cloud Services (with SPIKE)". Symposium on the Foundations of Software Engineering (ESEC/FSE 2019)

[4] Junjie Wang, Song Wang, **Jianfeng Chen**, Tim Menzies, Qiang Cui, Miao Xie and Qing Wang. "Characterizing Crowds to Better Optimize Worker Recommendation in Crowdsourced Testing.". IEEE Transactions on Software Engineering(2019).

[5] **Jianfeng Chen**, and Tim Menzies. "RIOT: A Stochastic-Based Method for Workflow Scheduling in the Cloud." 2018 IEEE 11th International Conference on Cloud Computing (CLOUD). IEEE, 2018. (Accept rate: 15%).

[6] Vivek Nair, Amrit Agrawal, **Jianfeng Chen**, Wei Fu, George Mathew, Tim Menzies, Leandro Minku, Markus Wagner, and Zhe Yu. "Data-Driven Search-based Software Engineering." The Mining Software Repositories (MSR) 2018.

[7] **Jianfeng Chen**, Vivek Nair, Rahul Krishna, and Tim Menzies. "Sampling as a Baseline Optimizer for Search-based Software Engineering." IEEE Transactions on Software Engineering (2018).

[8] **Jianfeng Chen**, Vivek Nair, and Tim Menzies. "Beyond evolutionary algorithms for search-based software engineering." Information and Software Technology (2017).

[9] Vivek Nair, Tim Menzies, and **Jianfeng Chen**. "An (accidental) exploration of alternatives to evolutionary algorithms for SBSE." In International Symposium on Search Based Software Engineering, pp. 96-111. Springer, Cham, 2016.

[10] Yijun Yang, Wei Zeng, and **Jianfeng Chen**. "Equiareal parameterizations of NURBS surfaces". Graphical models 76.1 (2014)